



Evaluating the impact of wildfire smoke on solar photovoltaic production

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Highlights

- Analyzed wildfire impact on solar PV production at 53 utility sites in Western US.
- Skilled random forest model predicts solar PV production with $R^2 = 0.91$ and $MSE = 0.005$.
- Wildfire smoke in vicinity reduced solar PV production by 8.3% on average.

Abstract

There are growing needs to understand how extreme weather events impact the electrical grid. Renewable energy sources such as solar photovoltaics are expanding in use to help sustainably meet electricity demands. Wildfires and, notably, the widespread smoke resulting from them, are one such extreme event that can impair the performance of solar photovoltaics. However, isolating the impact that smoke has on photovoltaic energy production, separate from ambient conditions, can be difficult. In this work, we seek to understand and quantify the impacts of wildfire smoke on solar photovoltaic production within the Western United States. Our analysis focuses on the construction of a random forest regression model to predict overall solar photovoltaic production. The model is used to separate and quantify the impacts of wildfire smoke in particular. To do so, we fuse historical weather, solar photovoltaic energy production, and PM2.5 particulate matter (primary smoke pollutant) data to train and test our model. The additional weather data allows us to capture interactions between wildfire smoke and other ambient conditions, as well as to create a more powerful predictive model capable of better quantifying the impacts of wildfire smoke on its own. We find that solar PV energy production decreases 8.3% on average during high smoke days at PV sites as compared to similar conditions without smoke present. This work allows us to improve our understanding of the potential impact on photovoltaic-

based energy production estimates due to wildfire events and can help inform grid and operational planning as solar photovoltaic penetration levels continue to grow.

Introduction

Wildfires are becoming both larger and more frequent across the United States (U.S.) [1]. Consequently, we are also seeing increasing damage and disruption to communities [2], livelihood [3], economies [4], [5], and local infrastructure such as transmission lines [6] and water supply systems [7]. While direct damage warrants much attention due to the vast destruction caused, wildfires also bring secondary effects in the form of smoke. The fine particulate matter in smoke leads to a reduction in air quality and visibility. Additionally, wildfire smoke often covers a vast area that is much larger than the actual burn area of the fire itself (see Figure A1 for satellite images from [8]). Many more communities are likely to be impacted by smoke than by the fire directly. The impacts to health and well-being are significant [9]. There are also adverse impacts to solar photovoltaic (PV) energy production, which can lead to additional stress for the power grid. Heavy smoke reduces the amount of solar radiation that can reach solar collection panels, and understanding the magnitude of energy production impacts can benefit grid planning and operational decision-making [10]. Although our analysis focuses on analyzing historical data from the Western United States, the impacts of wildfire smoke on solar PV energy production is relevant to global community as well. Beyond the U.S., wildfires are continuing to grow in frequency and severity [11], and solar PV deployment is also increasing [12].

Solar energy penetration levels are increasing in the U.S., particularly in some western states with high susceptibility and vulnerability to wildfire events. California, for example, generated over 15% of its in-state energy using solar power in 2020. During the same year, the state had 8,648 wildfire incidents with an estimated 17,419 square kilometers (4,304,379 acres) burned [13]. Smoke propagation from wildfires depends on wind speed, wind direction, terrain, and most critically, the burn patterns of the fires themselves [14]. Research has advanced significantly to better forecast wildfire smoke severity and transport, which can be used to provide advance warning of impacts to solar energy production [15]. However, the impact of smoke on solar PV production is still poorly understood. With a growing reliance on solar energy in grid operations, energy forecasts used for planning should incorporate smoke impacts during periods of projected wildfire activity and subsequent smoke propagation. Impacts to the grid may be short term (i.e., hours to days) from wildfire-related smoke. However, these trends can be explored to understand longer term PV production behavior, especially when sites may be exposed to multiple wildfire events over the course of their service lives. The value of accurate solar forecasts is already significant [16]. These insights will become even more important in a future where wildfire events are commonplace [17] and solar capacity on the grid increases.

Previous studies have demonstrated that pollution, and smoke in particular, can have a negative impact on the potential for solar PV production. Small PM_{2.5} particles (fine particulate matter of 2.5 μm or less in diameter) are the primary pollutant in wildfire smoke. It is well understood that wildfire emissions can spread dangerous levels of PM_{2.5} across a wide area, with some regions more prone to severe exposure than others [18], [19]. Focusing on solar PV production, several studies have relied on limited, experimental field data or laboratory testing [20], [21]. Other research efforts have used large, computational models with high levels of uncertainty [22], [23]. One study identified a negative impact to solar PV production associated with particulate matter from pollution sources [24]. These uncertainties make it difficult to use these models in an operational context when decisions need to be made about specific assets in the power grid. In several more targeted case studies, aerosols have been estimated to reduce utility-scale PV generation in Korea by 15%–24% using a regression analysis [25]. Experimental testing of performance during specific, highly localized wildfire events can produce estimates of both the average and instantaneous reduction in power production. Values for instantaneous reductions from smoke can be as high as 73% under extreme conditions [26]. In other cases, the maximum instantaneous impact from a well-characterized fire event was 27% [27]. While important to understand these localized impacts, there are many applications of grid planning for which larger scale and longer time frame estimates of impact are needed.

Several studies offer estimates for average reduction that fall in the range of 7%–30%, although the degree of impact can vary widely based on fire severity, proximity of a PV site, and atmospheric conditions [28], [29]. Keelin et al. quantify the potential adverse impact using a method that can be applied to any location, focusing on irradiance instead of measured production [30]. Juliano et al. [29] used pooled solar PV data provided by the California Independent System Operator. Their study spanned a 9-day period in 2020 and quantified a 10%–30% reduction in solar PV production at peak operating periods with additional variability in impact throughout the production day. Donaldson, Piper, and Jayaweera [31] used 10 sites in California and focused on a 59-day period during the 2020 wildfire season. Similar to Juliano et al. [29], their analysis was restricted to peak production hours versus using the full production day. They estimate a 9%–49% derating of solar PV production during wildfires, depending on the statistical model used. Multi-site assessments focused on peak production provide limited understanding of the total daily energy generation impact. Our analysis addresses this key gap by developing a model to understand the impact of wildfire smoke across the entire production day.

Our paper adds to the literature on understanding how wildfires can potentially impact solar PV energy generation. In particular, our work distinguishes itself from by Donaldson, Piper, and Jayaweera [31] and Juliano et al. [29] in multiple ways. Some key areas we are motivated to improve upon include: (1) the number of sites used (i.e., 53 sites versus 10); (2) consideration of multiple climate zones [32]; (3) the portion of the production day used to support analysis (i.e., full production day versus peak production hour); (4) the duration of wildfire events (i.e., 110 total wildfire days versus 9–59 days); and (5) usage of PM_{2.5} versus aerosol optical depth (AOD) to capture wildfire intensity.

Our objective is to provide a generalizable model that characterizes the role that PM_{2.5} plays in short term solar PV production while also accounting for the known factors that dictate production (e.g., cloud cover and wind speed). Through a case study of the Western United States, this enable us to answer our overarching question: What is the impact of wildfire smoke on solar PV energy production? To address this line of inquiry, we build on previous work aimed at characterizing this impact using measured data from 53 utility-scale solar PV sites across a period of time that captures smoke impacts from several wildfires, both large and small in terms of areal extent.

The Western U.S. experienced thousands of fires during the 2018 fire season [33]; it is a period of time with significant overlap with available solar PV energy production data. Three significant fire events were identified: the Ferguson Fire, the Mendocino Complex Fire, and the Camp Fire. The Ferguson Fire occurred in central California near Yosemite Valley where 392 square kilometers (96,901 acres) were burned from July 13 to August 19, 2018 [34]. The second major fire event was the Mendocino Complex Fire located in northern California. During this fire, 1,858 square kilometers (459,123 acres) burned from July 27 to September 18, 2018. It was the largest wildfire in California state history at the time [35]. The Camp Fire event occurred in Northern California where 620 square kilometers (153,336 acres) burned from November 8 to November 25, 2018. It resulted in the destruction of the town of Paradise, CA [36]. The 2018 wildfire season is consistent with a longer term trend of large scale fires (i.e., spanning at least 4.05 square kilometers) in the region [37].

The novelty of this work is the development of a machine learning-based model that isolates the effects from PM_{2.5} on solar PV energy production while controlling for other key drivers. Multiple months of jointly available multi-site solar PV production data and PM_{2.5} data adds to the model's robustness. The model can be used to assess the broad impact of wildfires historically at existing solar PV sites. Our work focuses on the effects of wildfire smoke and not their related soiling impacts as these are not possible to capture in within our dataset. Further, if coupled with forecast models, our technical approach can be used to predict expected near-term disruptions to solar energy production during active wildfire events.

The remainder of the paper is organized as follows. Section 2 provides an overview of the datasets used and describes our analytical approach. Section 3 presents the random forest model as well as single and multi-site analyses. Section 4 discusses limitations of the study and offers opportunities for future work. Section 5 concludes with a synthesis of the contributions and broader impact of this work. Additional supporting figures are included in the appendix.

Section snippets

Methods

The quantification of wildfire impacts on PV production is based on 53 sites in the U.S. Model development begins with the integration, processing, and evaluation of data from several sources, including solar PV production data, weather data, and PM2.5 data (see Fig. 1). We have combined these to create a comprehensive understanding of weather conditions and wildfire smoke pollution at each of the solar sites for which we have production data. Details of these data sources, the processing steps ...

Random forest model

Predicted solar PV energy production based on the LOO approach and the 20% testing data show comparable results (see Fig. 2). Notably, the 20% test result has a lower *MSE* (0.005 versus 0.007) and higher R^2 (0.91 versus 0.87) than the LOO model result. This is to be expected as the model used to predict on the 20% test data was trained on data from every site. The overall final model is able to explain 91% of the variability within the testing dataset with an error of approximately 0.5% of the...

Discussion

Our analysis is focused on quantifying impacts from wildfires at utility-scale PV plants located in the Western U.S. through the development of a machine learning-based predictive model. The final model developed has good modeling performance with a high R^2 (0.91) and low *MSE* (0.005). A multi-site impact analysis showed an average 8.3% reduction in solar PV performance due to wildfire-related smoke.

The analytical framework was designed to produce a generalizable model that could be applied to...

Conclusion

This study shows that the potential for production impacts across a large region subject to heavy smoke are significant. The development of a random forest-based model as well as the inclusion of synthetic PM2.5 data enabled us to isolate and quantify the impacts from smoke during known wildfire events. The RFM had leading features of clearness index and temperature at 2 m. PM2.5 (as a proxy for wildfire-related smoke) was of less overall importance. The isolation of PM2.5 through the usage of...

CRedit authorship contribution statement

Samuel D. Gilletly: Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing.

Nicole D. Jackson: Conceptualization, Project administration, Writing – original draft, Writing – review & editing.

Andrea Staid: Conceptualization, Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review & editing....

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper....

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